A Ka-Band CMOS Variable-Gain Phase Shifter With an Impedance-Invariant High-Gain Vector-Summing Amplifier

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***Abstract*—This article presents a Ka-band variable-gain phase shifter with low phase and gain errors. A current-reuse configuration with a transformer-coupled Gilbert cell (TCGC)-based vector-summing amplifier (VSA) is proposed and operated using current-steering digital-to-analog converters (DACs). The proposed VSA maintains constant input and output impedances across all gain and phase states and exhibits high-gain characteristics. This approach enables the phase shifter to achieve low root-mean-square (rms) errors and a high overall gain. Bleeding currents are also employed to tune the constellations of the phase shifter by controlling the gain of the *I* and *Q* paths and to mitigate load impedance variations in the VSA's common-source (CS) stage. Fabricated in a 40-nm CMOS technology, the phase shifter demonstrated a peak gain of 3.1 dB at 27.1 GHz and a 3-dB bandwidth of 25.3–29.6 GHz. At 28 GHz, the rms gain and phase errors were 0.1 dB and 0.58°, respectively, with 4-bit gain control over a 21-dB dynamic range and 6-bit phase control over 360°. Within the 3-dB bandwidth, the rms gain and phase errors were less than 0.2 dB and 0.75°, respectively. The core area and power consumption are 0.38 mm^2 and 11.3 mW, respectively.**

***Index Terms*—CMOS, high gain, impedance-invariant, Ka-band, variable-gain phase shifter, vector-summing amplifier (VSA).**

I. Introduction

RECENTLY, explorations of the millimeter-wave (mmWave) spectrum have gained significant attention, driven by applications such as 5G communications, satellite communications, and radar systems [1], [2], [3], [4], [5], [6], [7], [8]. Although mmWave operation presents challenges due to high atmospheric attenuation and significant path loss, which limit the transmission range, the short wavelength at mmWave frequencies enables the design of high-gain phased antenna arrays with a compact form factor [4], [5], [6]. Consequently, beamforming integrated circuits (ICs) have emerged as a key technology for mmWave applications.

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Phase and gain controls are essential in beamforming systems to achieve the desired beam shape and low sidelobe levels [7], [8], [9]. These functions are implemented using either passive or active phase shifters and variable gain amplifiers [10]. A passive phase shifter is typically implemented using a switched-line or reflective-type architecture, offering excellent linearity and zero power consumption [11], [12], [13]. However, it exhibits high insertion loss, necessitating additional amplifiers to achieve the desired gain. An active phase shifter is typically realized using a vector-sum architecture consisting of a quadrature generator and a vector-summing amplifier (VSA), which exhibits low insertion loss and readily achieves high resolution regardless of chip size [14], [15], [16], [17], [18]. Additionally, gain control functions can be easily integrated into the circuit [2], [19]. For these reasons, active phase shifters are better suited for high-performance RF beamforming systems. However, the VSA in an active phase shifter exhibits impedance variations across different phase and gain states, along with limited gain. These issues must be addressed, as they degrade the beam-shaping accuracy and power efficiency of the beamforming ICs. Several phase shifters based on complementary switched-type impedance-invariant VSAs have been proposed in the literature [10], [20], [21]. Although these VSAs effectively mitigate impedance variations, achieving high phase resolution requires a large MOS array, which increases both the overall array size and insertion loss.

In this article, a Ka-band variable-gain active phase shifter with high gain and low phase and gain errors is presented, utilizing a current-reuse configuration with the proposed transformer-coupled Gilbert cell (TCGC) VSA. Current-steering digital-to-analog converters (DACs) are used to control the currents of the VSA, which define the output vector. The proposed VSA maintains constant input and output impedances across all gain and phase states, thereby reducing the gain and phase errors of the phase shifter. Moreover, it enhances the gain without increasing the power consumption. Bleeding currents in the VSA are employed to mitigate load impedance variations in the common-source (CS) stage and to tune the overall constellation of the phase shifter.

The remainder of this article is organized as follows. Section II analyzes the Gilbert cell-based VSA and introduces the proposed impedance-invariant high-gain VSA. Section III

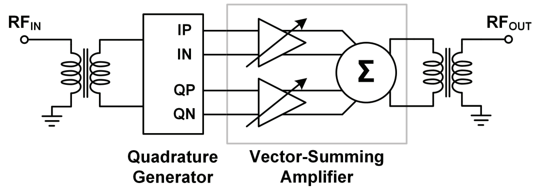


Fig. 1. Block diagram of a typical active vector-sum phase shifter.

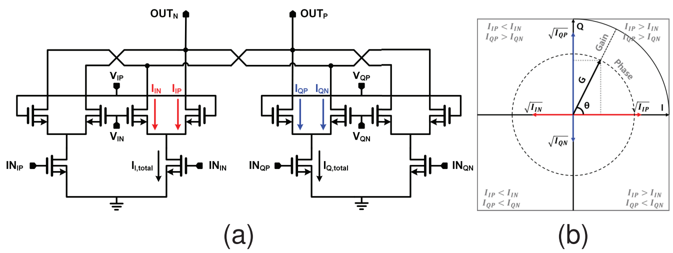


Fig. 2. (a) Schematic of the Gilbert cell-based VSA. (b) Synthesized output vector of the VSA.

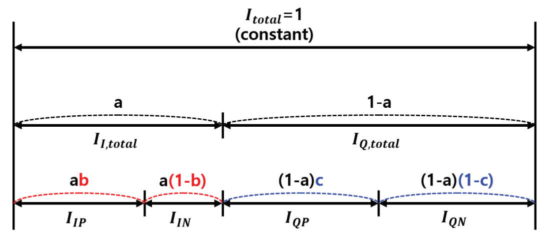


Fig. 3. Current flow diagram controlled by the current-steering DACs.

presents the detailed circuit implementation, and Section IV discusses the measurement results. Finally, Section V concludes this article.

II. Impedance-Invariant High-Gain VSA

A block diagram of an active vector-sum phase shifter is shown in Fig. 1, which typically consists of a quadrature generator and a VSA. The quadrature generator produces two orthogonal signals with a 90° phase difference, referred to as the I and Q signals. The VSA modulates the I and Q signals with the respective gain weight of each path and synthesizes the desired output vector by combining the two modulated signals. Because the VSA is positioned after the quadrature generator, its input impedance critically affects the accuracy of the quadrature signals. Additionally, the gain and power consumption of the VSA determine the overall efficiency of the phase shifter. Therefore, to realize a high-efficiency phase shifter with low phase and gain errors, an impedance-invariant high-gain VSA with low-power consumption is required. In this section, previously reported Gilbert cell-based VSAs and their limitations are discussed, and an improved VSA architecture is proposed.

A. Gilbert Cell-Based VSA

The Gilbert cell-based VSA is a widely used topology in active vector-sum phase shifters, even in the mmWave regime [14], [16], [19], [22], [23], [24]. This structure readily integrates variable-gain functionality to modulate the generated I and Q signals by adjusting the bias currents, which are controlled by either the first- or second-stacked transistors from the ground of the Gilbert cell-based VSA. In the former case [14], [19], the first-stacked transistors serve as tail current sources, and the RF signals are applied only to the second-stacked transistors. This results in a simple RF structure; however, it exhibits low gain due to the presence of only a single-stage RF amplification. In the latter case [16], [22], [23], [24], the RF signals are applied to the first-stacked transistors, and the second-stacked transistors are used to control the gains of the RF paths, as shown in Fig. 2(a). This configuration enables higher RF gain by incorporating an additional RF common-gate (CG) amplifier stage. Therefore, it is more advantageous for realizing a high-gain VSA. The output vector in Fig. 2(a) is synthesized by adjusting the gains of the CG stage, which are controlled by the bias currents I_{IP}, I_{IN}, I_{QP}, and I_{QN}, as shown in Fig. 2(b). The effective transconductances of the CG stage in the I and Q paths, $G_{m,I}$ and $G_{m,Q}$, can be expressed as follows:

$$G_{m,I} = k\sqrt{I_{IP}} - k\sqrt{I_{IN}} \tag{1}$$

$$G_{m,Q} = k\sqrt{I_{QP}} - k\sqrt{I_{QN}} \tag{2}$$

where

$$k = \sqrt{2\mu_{\text{eff}} C_{ox}\frac{W}{L}}. \tag{3}$$

The quadrant of the output vector is determined by the signs of $G_{m,I}$ and $G_{m,Q}$. In the first quadrant, where both $G_{m,I}$ and $G_{m,Q}$ are positive, the magnitude G and phase θ of the output vector are expressed as follows:

$$G \propto \sqrt{\left|G_{m,I}\right|^2 + \left|G_{m,Q}\right|^2} \tag{4}$$

$$\theta = \arctan\frac{\left|G_{m,Q}\right|}{\left|G_{m,I}\right|}. \tag{5}$$

The Gilbert cell-based VSA can synthesize the output vector simply by adjusting the bias current; however, it has certain drawbacks. First, the input and output impedances of the VSA vary depending on the phase and gain states of the output vector. These variations not only complicate impedance matching but also cause input impedance mismatches between the I and Q signal paths. Such variations and mismatches affect the quadrature generator, leading to amplitude and phase imbalances in the quadrature generator. These, in turn, increase the overall root-mean-square (rms) errors of the phase shifter. Therefore, the input and output impedances of the VSA should remain constant across all phase and gain states. Second, the maximum gain of the VSA is limited by the parasitic capacitance at the internode of the Gilbert cell. This reduces the overall gain of the phase shifter, necessitating the addition of a gain block. Finally, analog bias current control complicates the calibration of the phase shifter.

B. VSA With Current-Steering DACs

The output vector of the Gilbert cell-based VSA can be controlled by the specially designed current-steering DACs and double-pole double-throw (DPDT) switches for quadrant selection. The DACs determine the current ratio of I_{IP}, I_{IN}, I_{QP}, and I_{QN} while maintaining the total dc current, I_{total}, constant, as shown in Fig. 3. In the first quadrant, G and θ are represented as follows:

$$G \propto \sqrt{\left|\sqrt{a}\left(\sqrt{b}-\sqrt{1-b}\right)\right|^2+\left|\sqrt{1-a}\left(\sqrt{c}-\sqrt{1-c}\right)\right|^2} \tag{6}$$

$$\theta = \arctan \frac{\left|\sqrt{1-a}\left(\sqrt{c}-\sqrt{1-c}\right)\right|}{\left|\sqrt{a}\left(\sqrt{b}-\sqrt{1-b}\right)\right|} \tag{7}$$

where a is the I-path current ratio with respect to I_{total}, b is the I_{IP} current ratio with respect to the I-path current, and c is the I_{QP} current ratio with respect to the Q-path current, respectively (a, b, c in [0, 1]). These current-steering DACs ensure a constant output impedance across all gain and phase states, thereby minimizing phase and gain errors. Additionally, the use of digital control codes simplifies gain and phase control.

By setting $a = 0.5$ to reduce the number of control bits in the DACs, G and θ can be expressed as follows:

$$G \propto \sqrt{\left|\sqrt{b}-\sqrt{1-b}\right|^2+\left|\sqrt{c}-\sqrt{1-c}\right|^2} \tag{8}$$

$$\theta = \arctan \frac{\left|\sqrt{c}-\sqrt{1-c}\right|}{\left|\sqrt{b}-\sqrt{1-b}\right|}. \tag{9}$$

The output vector can be represented in Cartesian coordinates, where G and θ are dependently determined. The currents through the input transistors of the CS stage remain equal and constant across all phase and gain states, significantly reducing input impedance variations. However, input impedance variations in the VSA still occur due to the parasitic gate-drain capacitance C_{gd}, which reflects load impedance variations in the CS stage across different states [16]. Given that the gain and phase of a channel in a phased-array system are typically represented in polar coordinates, phase shifters with constellations in Cartesian coordinates cannot fully utilize the four corner regions of the constellation. On the other hand, by setting $b = c$, G and θ can be expressed as follows:

$$G \propto \left|\sqrt{b}-\sqrt{1-b}\right| \tag{10}$$

$$\theta = \arctan \frac{\left|\sqrt{1-a}\right|}{\left|\sqrt{a}\right|}. \tag{11}$$

The output vector can be represented in polar coordinates, where G and θ are determined independently. This approach results in lower rms errors and higher control efficiency compared to Cartesian coordinate control in a phased-array system. Additionally, polar coordinate control provides a higher peak available gain for the same I_{total}.

The Gilbert cell-based VSA with current-steering DACs consistently exhibits input impedance variations and mismatches across all states. Additionally, the load impedance variations in the CS stage hinder the precise control of the VSA, which also becomes another reason for the input impedance variations. These issues are exacerbated in polar coordinate control, which has advantages in phased-array systems. Therefore, a VSA architecture is required that ensures constant input and output impedances, regardless of the control method.

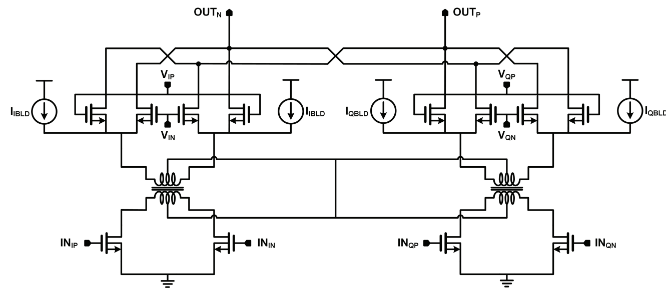


Fig. 4. High-gain VSA architecture with constant input and output impedances.

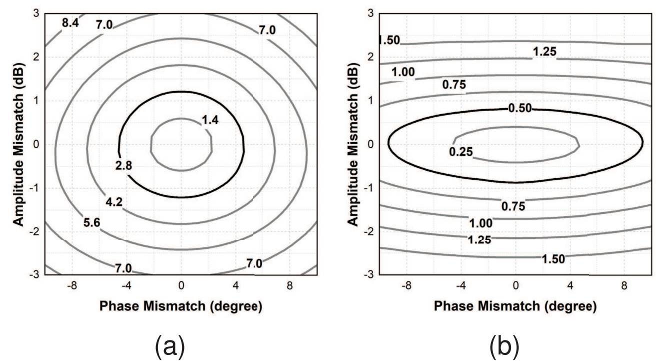
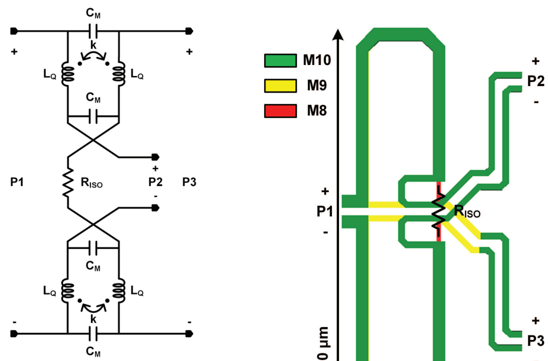


Fig. 5. Contours of phase and gain accuracy degradations in the phase shifter with a 6-bit phase resolution due to ΔA and $\Delta\phi$. (a) Calculated rms phase error. (b) Calculated rms gain error.

C. Proposed VSA Architecture

An impedance-invariant high-gain VSA architecture is proposed to address the aforementioned issues. It comprises the TCGC VSA, which incorporates a current-reuse configuration and bleeding currents, as shown in Fig. 4. The proposed architecture is controlled by current-steering DACs and DPDT switches to maintain a constant I_{total} during gain and phase control.

The transformers are employed at the interstages of the Gilbert cells to enhance the gain of the VSA by resonating out parasitic capacitance. Although either a parallel or series inductor can be used to cancel parasitic capacitance, combining both inductors is more effective for increasing the gain. However, directly combining the two inductors results in a significant area overhead. For this reason, a transformer is employed to replace them within the footprint of a single inductor. Note that the simplified equivalent model of the transformer can be regarded as a combination of series and parallel inductors. A current-reuse configuration is implemented by connecting the center taps of the transformers, establishing identical biasing conditions for the transistors in the CS stage. This configuration enforces an equal split of I_{total}



L_Q	k	C_M	R_{ISO}
180 pH	0.85	120 fF	50 Ω

Fig. 6. (a) Schematic and (b) layout of the fully differential hybrid coupler.

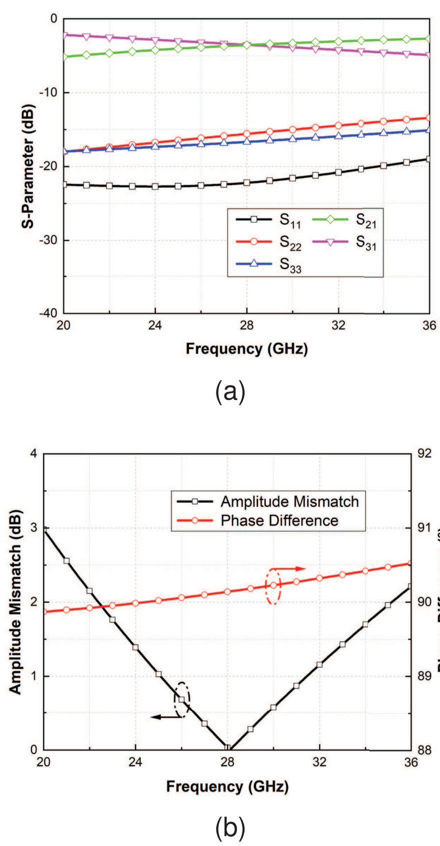


Fig. 7. Simulation results of the fully differential hybrid coupler: (a) S-parameters and (b) amplitude mismatch and phase difference.

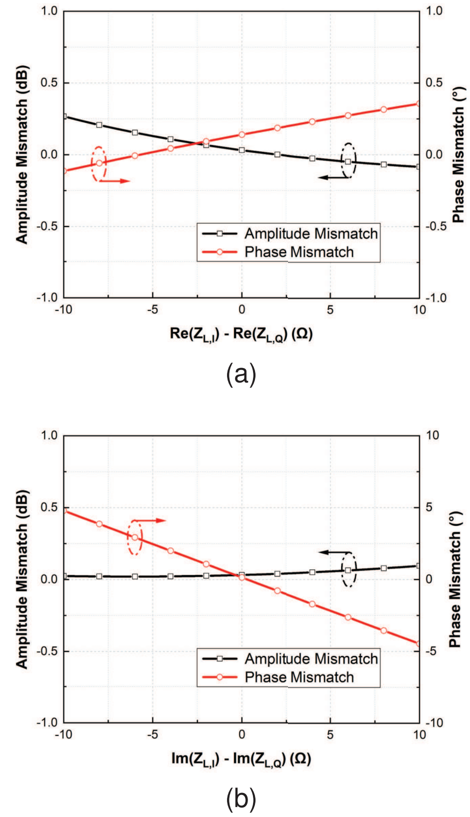


Fig. 8. Amplitude and phase mismatches of the differential hybrid coupler with impedance mismatches in the I and Q loads at 28 GHz: (a) with respect to the real part mismatch and (b) with respect to the imaginary part mismatch.

among the input transistors, regardless of the control method of the current-steering DACs for the Cartesian (a = 0.5) or polar ($b = c$) constellations. It ensures that the input transistors operate exclusively in the saturation region and significantly reduces variations in the input impedance.

The bleeding currents in the I and Q paths of the CG stage are introduced to mitigate load impedance variations in the CS stage across different gain and phase states. These bleeding currents further minimize input impedance variations and enhance the control accuracy of the VSA. They can also be utilized to control the gain of the I and Q paths in the VSA. To increase the I-path gain relative to the Q-path gain, the bleeding current in the Q path should be larger than that in the I path. Conversely, to increase the Q-path gain relative to the I-path gain, the bleeding current in the I path should be larger than that in the Q path. Note that the total sum of the bleeding currents is maintained to keep the total current I_{total} constant. This tunability is leveraged to calibrate the distorted constellations of the phase shifter caused by the quadrature generator or the nonideal effects of the chip. Therefore, the proposed VSA architecture maintains constant input and output impedances regardless of the control states, while exhibiting high gain and enabling tunability of the overall phase shifter constellation.

III. Circuit Implementation

The variable-gain phase shifter consists of a differential hybrid coupler, an improved TCGC VSA with a current-reuse configuration controlled by current-steering DACs, and impedance-matching networks. The impedance-matching networks are implemented using transformer-based configurations to achieve wideband impedance matching.

A. Quadrature Generator

The amplitude and phase mismatches of the quadrature generator directly affect the gain and phase errors of the

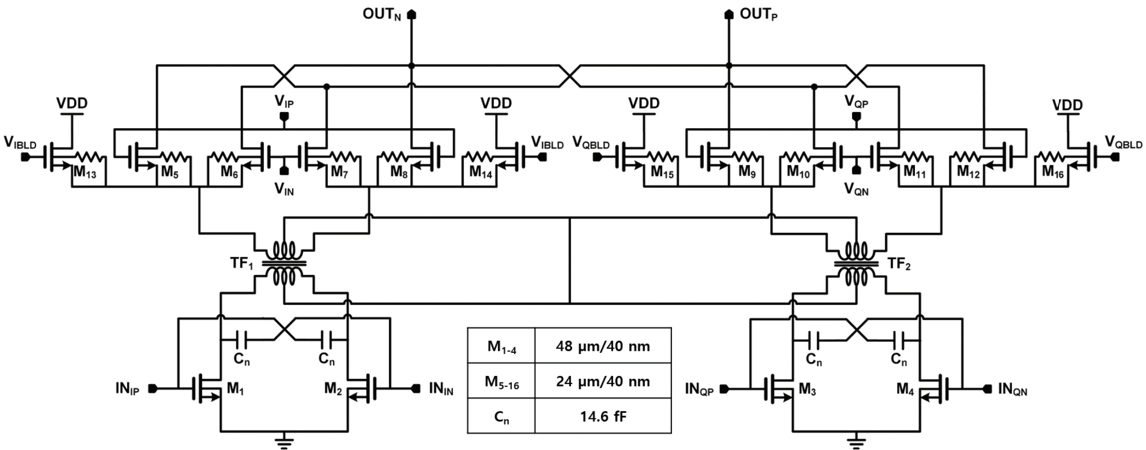


Fig. 9. Schematic of the proposed VSA.

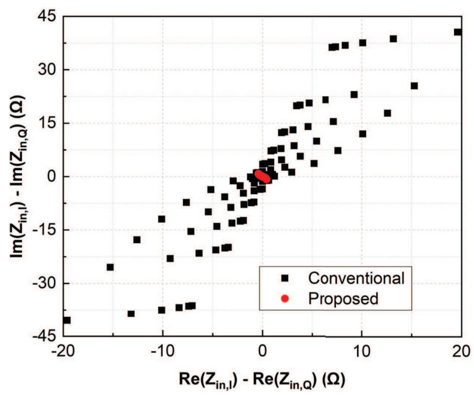


Fig. 10. Simulated input impedance mismatches between the I and Q paths of the VSA across all gain and phase states at 28 GHz.

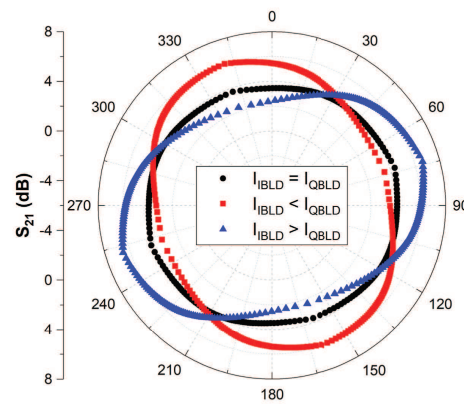
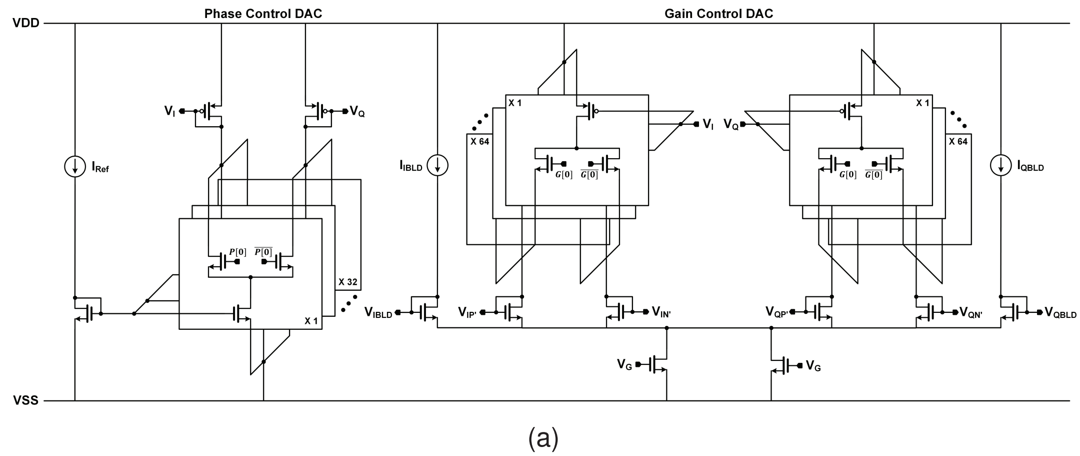


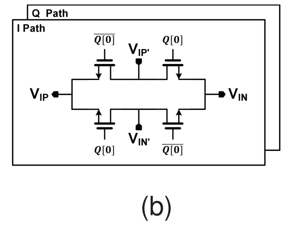
Fig. 11. Simulated constellations at the maximum gain state of the phase shifter as a function of the bleeding current magnitudes in the I and Q paths of the VSA.

phase shifter. For simplicity, it is assumed that the amplitude and phase mismatches, denoted as ΔA and $\Delta\phi$, respectively, represent the amplitude and phase mismatches in the Q signal relative to the ideal I signal. Fig. 5 shows the calculated rms phase error and rms gain error in a phase shifter with a 6-bit phase resolution, assuming an ideal gain weighting in the VSA. The rms phase error should be less than 2.8125°, as the 6-bit phase resolution corresponds to 5.625°. Therefore, the ΔA and $\Delta\phi$ should be within the 2.8° contour in Fig. 5 (a). To avoid the use of additional circuits for compensating gain variations between the phase states, the maximum rms error is set to less than 0.5 dB in this study; thereby, ΔA and $\Delta\phi$ should be within the 0.5 dB contour in Fig. 5(b). The target values for ΔA and $\Delta\phi$ are less than 0.8 dB and 2°, respectively, to satisfy these conditions.

A hybrid coupler or a polyphase filter is generally used to generate quadrature signals in the mmWave frequency band [10], [21], [25], [26]. A hybrid coupler is chosen in this design due to its significantly lower insertion loss compared to that of a polyphase filter. A fully differential structure is employed to mitigate the effects of imperfect grounding [21], [25], [26]. Fig. 6(a) shows a schematic of the fully differential hybrid coupler. The reference impedance is set to a differential 50 Ω to reduce the chip area [27]; thus, the isolation resistor R_{ISO} is set to 50 Ω. Fig. 6(b) shows the layout of the implemented coupler, which occupies an area of 0.047 mm^2. Simulation results of the designed hybrid coupler are shown in Fig. 7. Both the I and Q paths exhibit an insertion loss of −3.5 dB at 28 GHz, including the intrinsic 3 dB loss due to the I/Q division. The amplitude and phase mismatches are less than 0.69 dB and 0.23°, respectively, within the 26–30 GHz band, which is sufficient to achieve a 6-bit phase resolution with an rms gain error of less than 0.5 dB.

Fig. 8 shows the simulated amplitude and phase errors of the coupler caused by load impedance mismatches in the I and Q paths at 28 GHz. The default values of the I and Q load impedances, $Z_{L,I}$ and $Z_{L,Q}$, are 50 + j0 Ω. Based on these values, the real and imaginary components of $Z_{L,Q}$ are varied from 40 to 60 Ω and from −10 to 10 Ω, respectively. Fig. 8(a) shows the errors due to the real part mismatch when the imaginary part mismatch is 0 Ω. The amplitude and phase mismatches increase by 0.36 dB and 0.27°, respectively. Fig. 8(b) shows the results for the imaginary part mismatch when the real part mismatch is 0 Ω. The amplitude and phase mismatches increase by 0.09 dB and 4.79°, respectively. The





Q[0]	Q[1]	Quadrant
0	0	1
1	0	2
1	1	3
0	1	4

(c)

Fig. 12. Schematics of (a) current-steering DACs for phase and gain control, (b) quadrant selection switches, and (c) quadrant selection logic.

amplitude error is more influenced by the real part mismatch, while the phase error is more influenced by the imaginary part mismatch. In a practical circuit, as both the real and imaginary impedance mismatches occur simultaneously, the errors of the quadrature generator will be exacerbated.

B. Vector-Summing Amplifier

A schematic of the VSA based on the architecture proposed in Section II is shown in Fig. 9. The interstage transformers TF_1 and TF_2 have identical design values, which are determined to achieve impedance matching. A current-reuse configuration is implemented by connecting the center taps of the transformers using thick metal. The bleeding currents are implemented using transistors with dimensions identical to those in the CG stage. In addition, a capacitive neutralization technique is applied to cancel C_{gd}, resulting in a constant input impedance. This is enabled by the fact that the input transistors operate exclusively in the saturation region due to the current-reuse configuration. The bodies of the transistors M_{5-16} are floated through large resistors to reduce the parasitic drain-to-source capacitance C_{ds}.

Fig. 10 shows the simulated input impedance mismatches between the I and Q paths of the VSA across all gain and phase states at 28 GHz, where $Z_{in,I}$ and $Z_{in,Q}$ denote the input impedances of the I and Q paths, respectively. The x-axis and y-axis represent the mismatch in the real and imaginary parts, respectively. Impedance mismatches of 19.6 and 40.6 Ω in the real and imaginary parts, respectively, are observed in the conventional Gilbert cell-based VSA. In contrast, the proposed VSA significantly reduces these mismatches to 0.4 and 0.8 Ω in the real and imaginary parts, respectively. These results demonstrate that the proposed VSA effectively mitigates impedance variation and mismatch issues in the I and Q paths.

Fig. 11 shows the simulated constellations in the maximum gain state of the phase shifter as a function of the bleeding current magnitudes along the I and Q paths. If the constellations are distorted by the nonideal effects of the chip or the amplitude mismatches in the quadrature generator, the bleeding currents can be used to tune the outer boundary of the constellations and increase the peak available gain. This can reduce the rms errors and increase the operational bandwidth of the phase shifter.

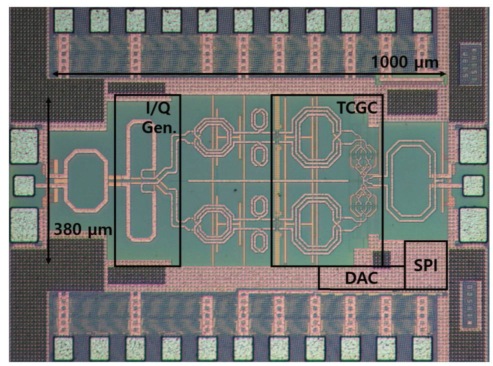


Fig. 13. Chip photograph of the variable-gain active phase shifter.

A schematic of the specially designed current-steering DACs for the selection of the phase and gain is shown in Fig. 12(a). The parameter a in Fig. 3 is determined using 6-bit binary-weighted control, while b, which is equal to c, is determined using 7-bit binary-weighted control. The output control voltages of the DACs are set through the complementary operation of the single-pole double-throw (SPDT) switches. A pair of DPDT switches, as shown in Fig. 12(b), determines the quadrant of the output vector, and Fig. 12(c) illustrates the corresponding quadrant selection logic.

IV. Measurement Results

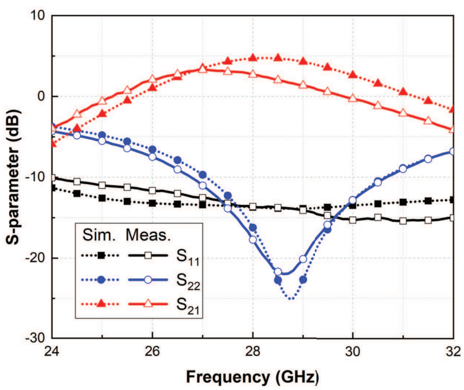
The variable-gain active phase shifter was implemented using a 40-nm CMOS process with a core area of 0.38 mm^2, excluding pads, as shown in Fig. 13. The device under test (DUT), mounted on an FR4 PCB, was measured via on-wafer probing using ground–signal–ground (GSG) probes. The total power consumption, including the current-steering DACs, was 11.3 mW for a supply voltage of 1.1 V. The S-parameters were measured using an Anritsu 37397D vector network analyzer. All gain and phase states were controlled using a Cheetah SPI adapter, and the measured data were acquired via a general-purpose interface bus (GPIB) cable.

TABLE I
PERFORMANCE SUMMARY AND COMPARISON

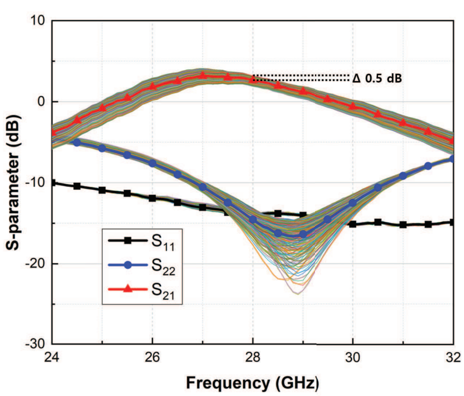
	This work	[15]	[16]	[17]	[18]	[13]
Process	40-nm CMOS	65-nm CMOS	28-nm CMOS	65-nm CMOS	65-nm CMOS	65-nm CMOS
Topology	Active	Active	Active	Active	Active	Passive
Frequency (GHz)	25.3-29.6	25.4-35.6	21.7-28.6	25-30	30-32.5	32-40
Peak Gain (dB)	3.1	0.6	0.6	-8	-2.8	-16.9
Phase Resolution (bits)	6	4	6	6	4	7
Gain Dynamic Range (dB)	21	18	10	3	17.3	-
Gain Resolution (bits)	4	3	4	2	2	-
RMS Phase Error (°)	< 0.75	< 3	< 1.5	< 3.55	< 3.5	< 1.6
RMS Gain Error (dB)	< 0.2	< 0.37	< 0.25	< 0.54	< 0.4	< 0.38
Power Consumption (mW)	11.3	40	14.4	38	18	0
Core Area (mm^2)	0.38	0.19	0.47	0.13	0.21	0.14
FoM1	1301.8	396.6	715.5	88.5	63.3	172.3
FoM2	5170.2	236.3	628.5	6.6	51.5	-

$$\text{FoM1} = \frac{f_0(\text{GHz}) \times G_{Pk}(\text{lin.}) \times BW_{3dB}(\text{GHz}) \times \theta_{Res.}(\text{bits})}{\theta_{RMS,3dB}(\text{deg.}) \times G_{RMS,3dB}(\text{lin.})}$$

$$\text{FoM2} = \frac{f_0(\text{GHz}) \times G_{Pk}(\text{lin.}) \times BW_{3dB}(\text{GHz}) \times \theta_{Res.}(\text{bits}) \times G_{Res.}(\text{bits}) \times G_{DR}(\text{lin.})}{\theta_{RMS,3dB}(\text{deg.}) \times G_{RMS,3dB}(\text{lin.}) \times P_{DC}(\text{mW})}$$

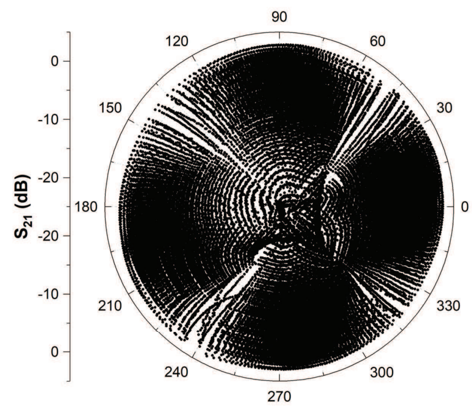


(a)

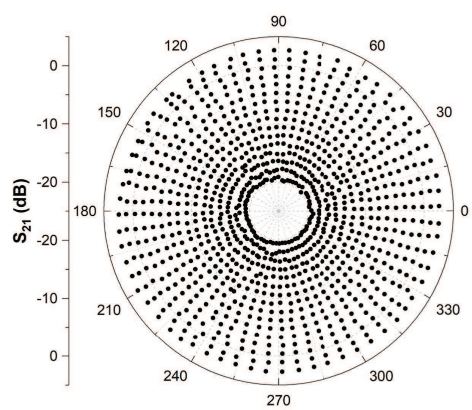


(b)

Fig. 14. (a) Simulated and measured S-parameters at the 0° phase state and (b) measured average S-parameters in 256 maximum gain states.



(a)



(b)

Fig. 15. Measured constellations at 28 GHz for (a) all gain and phase states and (b) 4-bit gain and 6-bit phase states.

The simulated and measured S-parameters of the phase shifter in the 0° states are shown in Fig. 14(a). The measured input and output reflection coefficients were less than −10 dB in the operating frequency range and closely matched the simulation results. Fig. 14(b) presents the measured average S-parameters of the phase shifter in the maximum gain state. The implemented phase shifter demonstrated constant input and output reflection coefficients with minor fluctuations. The measured average peak gain was 3.1 dB at 27.1 GHz, with

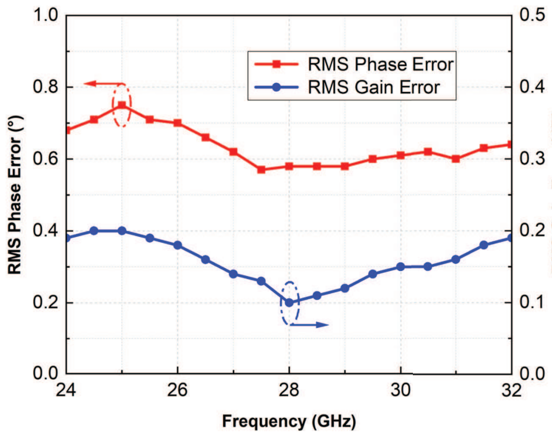


Fig. 16. Measured rms gain and phase errors for 4-bit gain and 6-bit phase states.

a 3-dB bandwidth of 25.3–29.6 GHz. The gain variation was less than 0.5 dB at 28 GHz. Fig. 15(a) shows the measured constellations across all gain and phase states using all control bits, and Fig. 15(b) presents the measured constellations with 6-bit phase control over 360° and 4-bit gain control over a 21-dB dynamic range at 28 GHz, selected using the lookup table-based calibration algorithm [26]. The calculated rms gain and phase errors were 0.1 dB and 0.58°, respectively. In the frequency range of the 3-dB bandwidth, the total rms gain and phase errors were less than 0.2 dB and 0.75°, respectively, as shown in Fig. 16. Note that these small-signal measurements were achieved without tuning using the bleeding currents in the VSA.

A performance comparison with previously reported phase shifters is summarized in Table I. Two figures of merit (FoMs) are used: FoM1, from [28], and FoM2, newly defined based on FoM1 to evaluate the variable-gain functionality. The proposed phase shifter exhibits the highest maximum gain with low-power consumption and the lowest rms phase and gain errors. These results demonstrate that the presented phase shifter effectively reduces rms errors and increases the gain through the use of the proposed high-gain VSA with constant input and output impedances.

V. Conclusion

A TCGC-based VSA incorporating a current-reuse configuration and bleeding currents is proposed to reduce gain and phase errors and to increase the overall gain of a variable-gain phase shifter. Transformers in the VSA serve as interstage matching elements, enhancing the gain without additional power consumption. A current-reuse configuration, achieved by connecting the center taps of the interstage transformers, is introduced to ensure constant input impedance. Bleeding currents are employed to adjust the constellations of the phase shifter by controlling the gain of the *I* and *Q* paths while mitigating load impedance variations in the CS stage of the VSA. The phase shifter achieves a peak available gain of 3.1 dB with a power consumption of 11.3 mW. The total rms gain and phase errors were less than 0.2 dB and 0.75°, respectively. To the best of the authors' knowledge, these results represent the highest peak gain per unit power and the lowest rms errors among previously reported phase shifters.

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